Unsteady state material balances on a CVD reactor worksheet

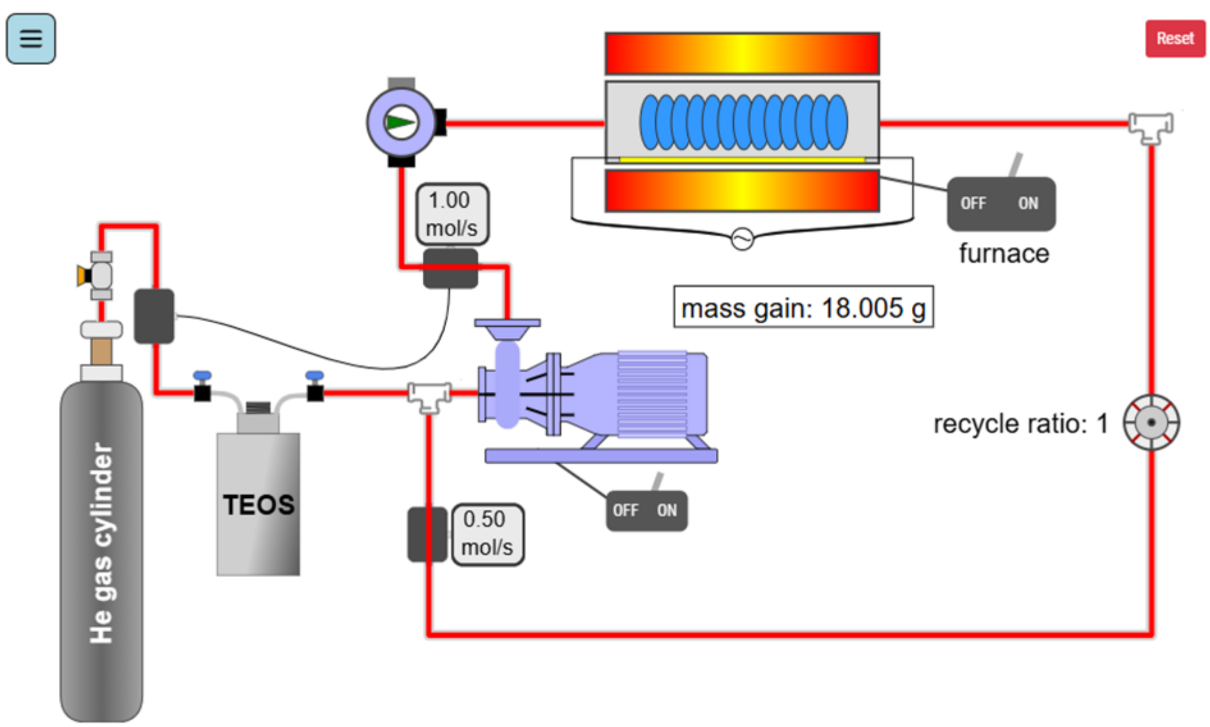
Name(s): ____

In this experiment, a stream of inert helium containing 0.34 mol% of the reactant tetraethyl orthosilicate (TEOS, $Si(C_2H_5O)_4$) is fed into a high-temperature reactor, where it produces solid-phase SiO_2 and vapor-phase water and ethylene. Silicon dioxide (SiO_2) deposits on material surfaces inside the chemical vapor deposition (CVD) reactor. The reactor has a recycle stream. Material balances are calculated for the system.

Student Learning Objectives

1. Write transient material balances for a reacting system with accumulation terms.
2. Relate mass deposition to reaction extent.
3. Perform balances around a reactor, mixer, and splitter.
4. Analyze the effect of recycle ratio on flow rates and conversion.

Equipment



The quartz crystal microbalance (QCM) measures the mass gain by the samples because of SiO_2 film deposition. The reaction is:

$$Si(C_2H_5O)_4(v) \rightarrow SiO_2(s) + 4\ C_2H_4(g) + 2\ H_2O(g)$$

Only a portion of the TEOS reacts. The stream exiting the reactor flows out of the process unless a valve is clicked open and a recycle ratio R (defined below) is set so that a portion of the exit stream from the reactor mixes with the fresh feed stream. The total flow to the reactor is maintained constant at 1.00 mol/s. Thus, when the recycle ratio changes, the feed from the TEOS bubbler changes because of feedback from the mass flow meter at the pump outlet.

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Notation

R = recycle ratio = $\frac{recycle\ flow\ rate}{fresh\ feed\ flow\ rate}$

$\dot{n}^{bubbler}_{TEOS_{no_{recycle}}}$ = TEOS molar flow rate from the bubbler when R – 0

$\dot{n}^{bubbler}_{TEOS}$ = TEOS molar flow rate from the TEOS bubbler

$\dot{n}^{recycle}_{TEOS}$ = TEOS molar flow rate in recycle stream

$\dot{n}^{reactor-in}_{TEOS}$ = TEOS molar flow rate into the reactor

$\dot{n}^{reactor-out}_{TEOS}$ = TEOS molar flow rate out of the reactor

$\dot{n}^{system\ out}_{TEOS}$ = TEOS molar flow rate out of system into hood

$\dot{m}_{SiO_2}$ = mass deposition rate of SiO_2

Questions to answer before starting the experiment

1. How is the mass gain of the solid surface related to the number of moles of TEOS reacted?

2. Will the TEOS concentration entering the reactor increase or decrease with recycle? Explain.

3. If the flow rate processed by the pump is constant, how will increasing the recycle ratio affect the total flow rate of the inlet and exit streams?

4. How do you expect the deposition rate will change with recycle ratio?

5. Will the overall fraction of TEOS leaving the process increase or decrease with recycle?

Running the experiment:

1. The 3-way valve at the reactor inlet should be flowing to the hood. The recycle ratio should be set to zero.
2. Open the He gas cylinder by clicking on the valve.
3. Turn on the feed pump.
4. Click on the 3-way valve so the TEOS/He stream flows through the reactor instead of to the hood.
5. Turn on the reactor furnace.
6. Wait 20 s for the system to stabilize, then record the mass gain and start a timer. After sufficient time for accurate measurements, record the mass gain and the time in Table 1. Calculate the rate of mass gain and record in Table 1. Repeat the measurement and record the result in Table 1.
7. Click on the valve, set the recycle ratio to 1.0, and repeat the measurements and record in Table 1.
8. Repeat the measurements at recycle ratios of 2.0 and 5.0 and record in Table 1.
9. Calculate the average rate of mass gain at each recycle ratio and record in Table 1.

Table 1 Mass gain for CVD reactor

Recycle ratio	Initial mass gain (g)	Final mass gain (g)	Net mass gain (g)	Elapsed time	Rate of mass gain (g/s)	Average rate of mass gain (g/s)
0						
0						
1.0						
1.0						
2.0						
2.0						
5.0						
5.0						

10. What trend do you notice in mass gain as the recycle ratio increases?

After the experiment:

1. Analyze the system in the absence of recycle. Here, the total flow rates of the stream leaving the bubbler and the stream entering the reactor are identical. Compute the molar flow rates of TEOS at the reactor inlet and reactor outlet by conducting a material balance on the reactor using the experimental information on rates of mass gain in the reactor.

2. When recycle is introduced, the molar flow rate of TEOS into the entire system (from the bubbler) is **not** equal to the molar flow rate entering the reactor. To account for this, set up additional material balances at the split point and at the mixing point. The inlet molar flow rate to the reactor will thus include the TEOS molar flow rate from the bubbler plus the TEOS molar flow rate from the recycle stream. <u>The pump maintains a constant volumetric flow rate of the gas stream through the reactor</u>, which provides an additional constraint. Develop an expression for the molar flow rate out of the mixing point in terms of the recycle ratio R, the molar flow rate of TEOS out of the reactor, and the *original* molar flow rate of TEOS when R was set to zero.

$$\dot{n}_{TEOS}^{reactor-in} = \dot{n}_{TEOS}^{bubbler} + \dot{n}_{TEOS}^{recycle}$$

$$\dot{n}_{TEOS}^{reactor-in} = \frac{1}{R+1}\dot{n}_{TEOS\_no\_recycle}^{bubbler} + \frac{R}{R+1}\dot{n}_{TEOS}^{reactor-out}$$

where $\dot{n}_{TEOS\_no\_recycle}^{bubbler}$ is a constant value that represents the flow from the bubbler when R = 0. When you specify that value and R, the two unknowns are $\dot{n}_{TEOS}^{reactor-in}$ and $\dot{n}_{TEOS}^{reactor-out}$.

3. In this system, the pump maintains a constant volumetric flow rate of the gas stream. At constant temperature, the outlet molar flow rate of TEOS can be modeled as a constant fraction of the inlet molar flow rate of TEOS. In other words, the reaction is expected to follow *first-order kinetics*. Use the mole balance from part 2 together with a material balance around the reactor (which should provide two equations with two unknowns) to complete Table 2. Does the system exhibit first-order kinetics?

Table 2 Molar flow rates

Recycle ratio	$\dot{n}_{TEOS}^{reactor-in}$ (mol/s)	$\dot{n}_{TEOS}^{reactor-out}$ (mol/s)	$\frac{\dot{n}_{TEOS}^{reactor-in}}{\dot{n}_{TEOS}^{reactor-out}}$
0.0			
1.0			
2.0			
5.0			

4. Complete a mole balance on the entire process and use it to complete Table 3.

Table 3 TEOS molar flow rates into and out of system

Recycle ratio	$\dot{n}_{TEOS}^{bubbler}$ (mol/s)	$\dot{n}_{TEOS}^{system\ out}$ (mol/s)	$\frac{\dot{n}_{TEOS}^{bubbler}}{\dot{n}_{TEOS}^{system\ out}}$
0.0			
1.0			
2.0			
5.0			

Are these ratios constant? Why do you think they have the trend that they do?